Review Test Submission: TSP 02

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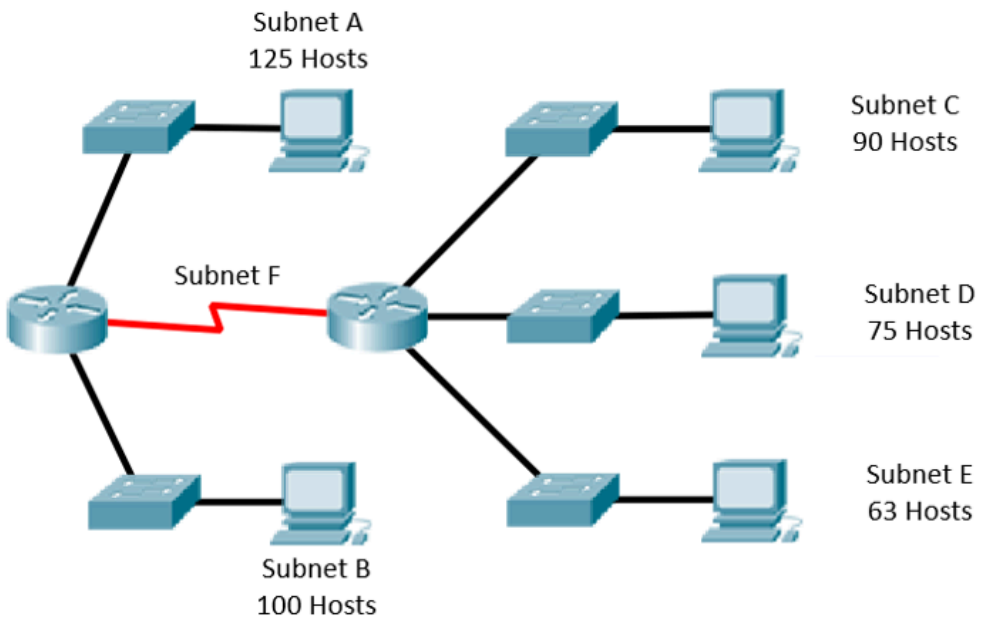
Course	Fall 2024 Switching and Routing (NET-126-2001)
Test	TSP 02
Started	11/7/24 11:07 AM
Submitted	11/7/24 11:18 AM
Status	Completed
Attempt Score	100 out of 100 points
Time Elapsed	11 minutes out of 2 hours
Results Displayed	Submitted Answers, Incorrectly Answered Questions

Question 1

4 out of 4 points



Refer to the topology. Use traditional subnetting to develop an addressing scheme based on the number of **hosts** using 172.16.0.0/16 as the starting point, then answer all of the questions below. Include the prefix length for the network addresses only. Express the interval and octet using the format: 32, 4th

Determine the following: **Custom Address Map (CAM)**

Hint: Since the requirements are based on the number of hosts:

1. Use formula for Total Usable Addresses: $2^h - 2$
2. Set it greater than or equal to the largest network.
3. Solve for h.
4. The value of h represents the number of 'h' (host) bits in the CAM.

Selected Answer: N.N.ssssssss.shhhhhhh

Question 2 4 out of 4 points

Determine the following: **Interval and Transition Octet**

Record your answer using the format *interval, octet*. For example: 8, 2nd

Selected Answer: 128, 4th

Question 3 3 out of 3 points

Determine the following: **Prefix Length**

Specify it as /NN, for example: /12

Selected Answer: /25

Question 4 3 out of 3 points

Determine the following: **Custom Subnet Mask**

Selected Answer: 255.255.255.128

Question 5 4 out of 4 points

Determine the following: **Wildcard Mask**

Selected Answer: 0.0.0.127

Question 6 3 out of 3 points

Determine the following: **Number of Subnets Possible**

Selected Answer: 512

Question 7 3 out of 3 points

Determine the following: **Total Addresses per Subnet**

Selected Answer: 128

Question 8 4 out of 4 points

Determine the following: **Total Usable Addresses per Subnet**

Selected Answer: 126

Question 9 3 out of 3 points

Subnet A: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Hint 3: To determine the next network address:

- Add the interval to the transition octet of this network address.
- Note: The host bits of netwok addresses are always set to 0.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 172.16.0.0/25

Question 10 3 out of 3 points

Subnet A: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 172.16.0.1

Question 11 3 out of 3 points

Subnet A: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 172.16.0.126

Question 12 3 out of 3 points

Subnet A: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.0.127

Question 13 3 out of 3 points

Subnet B: Network Address and Prefix Length

Hint 1: Add the interval to the transition octet of the previous network address and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 172.16.0.128/25

Question 14 3 out of 3 points

Subnet B: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 172.16.0.129

Question 15 3 out of 3 points

Subnet B: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 172.16.0.254

Question 16 3 out of 3 points

Subnet B: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.0.255

Question 17 3 out of 3 points

Subnet C: Network Address and Prefix Length

Hint 1: Add the interval to the transition octet of the previous network address and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 172.16.1.0/25

Question 18 3 out of 3 points

Subnet C: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 172.16.1.1

Question 19 3 out of 3 points

Subnet C: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 172.16.1.126

Question 20 3 out of 3 points

Subnet C: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.1.127

Question 21 3 out of 3 points

Subnet D: Network Address and Prefix Length

Hint 1: Add the interval to the transition octet of the previous network address and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 172.16.1.128/25

Question 22 3 out of 3 points

Subnet D: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 172.16.1.129

Question 23 3 out of 3 points

Subnet D: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 172.16.1.254

Question 24 3 out of 3 points

Subnet D: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.1.255

Question 25 3 out of 3 points

Subnet E: Network Address and Prefix Length

Hint: Add the Interval to the Transition Octet of the previous network address and apply the proper prefix.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 172.16.2.0/25

Question 26 3 out of 3 points

Subnet E: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 172.16.2.1

Question 27 3 out of 3 points

Subnet E: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 172.16.2.126

Question 28 3 out of 3 points

Subnet E: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.2.127

Question 29 3 out of 3 points

Subnet F: Network Address and Prefix Length

Hint: Add the Interval to the Transition Octet of the previous network address and apply the proper prefix.

Record your answer using the following format as an example: 172.16.8.0/20

Selected Answer: 172.16.2.128/25

Question 30 3 out of 3 points

Subnet F: First Usable Address

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix length.

Selected Answer: 172.16.2.129

Question 31

3 out of 3 points

Subnet F: Last Usable Address

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix length.

Selected Answer: 172.16.2.254

Question 32

3 out of 3 points

Subnet F: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.2.255

Thursday, November 7, 2024 11:18:52 AM EST

← **OK**